Ch 02: `print()` Output, Comments and Variables

Application Program Development

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Designing a Program (1 of 2)

Program Development Cycle

1. **Design the Program**
   - Design is the most important part of program development
   - Understand the task that the program is to perform
2. **Write the Code**
   - Use a programming language, like Python
3. **Correct Syntax Errors**
4. **Test the Program**
   - A *logic error* is a mistake that does not prevent program from running, but causes incorrect results.
   - Unit testing, module testing, system testing, stress testing
5. **Correct Logic Errors**
   - *debug* find and correct logic errors

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Designing a Program (2 of 2)

- Design is the most important part of program development
- Understand task that the program is to perform
  - Work with customers and users
  - Create software requirements
- Determine the steps that must be taken to solve the problem
  - Break down required task into series of steps
  - Create an algorithm, listing logical steps that must be taken
- **Algorithm**: set of well-defined logical steps that must be taken to perform a task.

Pseudocode and Flowcharts

Pseudocode: fake code, informal, no syntax rules

```
Input the hours worked

Input the hourly pay rate

Calculate gross pay as hours worked multiplied by pay rate

Display the gross pay
```

Flowchart: diagram graphically depicting steps in a program

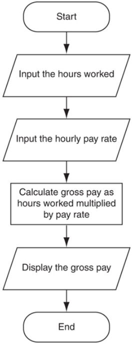

Figure 1: Flowchart for a pay calculating program (Gaddis, 2021, pg.57)

Input, Processing and Output

- Typically application performs three-step process
  1. Receive input
     - Input: any data that the program receives while it is running
  2. Perform some process on the input
     - Example: mathematical calculation
  3. Produce output

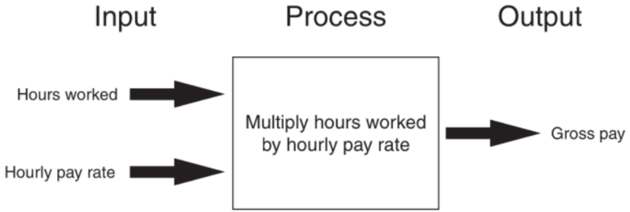

Figure 2: The input, processing, and output of the pay calculating program (Gaddis, 2021, pg. 58)

Displaying Output with the `print()` Function

- **Function**: piece of prewritten **reusable** code that performs an operation
- `print()` function: displays output on standard out (terminal)
- **Argument**: data given as input to a function
  - Example: give a string to be printed on terminal
- Statements in a program execute in the order that they appear (sequential execution)

```
c:\Users\etamustudent> python3
>>> print('Hello world')
Hello world
>>>
```

Strings and String Literals

- **String**: sequence of characters that is used as data
- **String literal**: string that appears in actual code of a program
  - Must be enclosed in single (') or double (“) quote
  - String literal can be inclosed in triple quotes (“ “ “) multiple lines

```
>>> print('Kate Austen')
Kate Austen
>>> print("123 Full Circle Drive")
123 Full Circle Drive
>>> contact = """
... Kate Austen
... 123 Full Circle Drive
... Asheville, NC 28899
... """
>>> print(contact)

Kate Austen
123 Full Circle Drive
Asheville, NC 28899
```

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Comments

- **Comments**: notes of (human readable, English sentences) explanation within a program
  - Ignored by Python interpreter
  - Intended to communicate information to code readers and maintainers
  - Begin with a # character
- **End-line comment**: appears at the end of a line of code
  - Typically explains the purpose of that line
  - Considered bad style in many cases, do not use

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Variables

- **Variable**: name that represents a value stored in the computer memory
  - Used to access and manipulate data stored in memory
  - A variable references the value it currently holds
- **Assignment statement**: used to create a variable and make it reference data
  - General format `variable = expression`
- In assignment statement, variable receiving value must be on left side
- A variable can be passed as an argument to a function
  - Variable name should not be enclosed in quote marks
- You can only use a variable if a value is assigned to it.

Variable Naming Rules

Rules for naming variables in Python:

- Variable name cannot be a Python key word
- Variable name cannot contain spaces
- First character must be a letter or underscore (not a number)
- After first character, may use letters, digits, or underscores
- Variable names are case sensitive

Variable name should reflect its use Meaningful Variable Names

Displaying Multiple Items with the print() Function

Python allows one to display multiple items with a single call to print()

- Items are separated by commas when passed as arguments
- Arguments displayed in the order they are passed to the function
- Items are automatically separated by a space when displayed on screen

```
>>> room = 503
>>> print('I am staying in room number', room)
I am staying in room number 503
```

Variable Reassignment

- Variables can reference different values while program is running
- **Garbage collection**: removal of values that are no longer referenced by variables
  - Carried out by Python interpreter
- A variable can refer to item of any type
  - Variable that has been assigned to one type can be reassigned to another type

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Numeric Data Types, Literals and the `str` Data Type

- **Data types**: categorize value in memory
  - e.g. `int` for integer `float` for real number, `str` used for storing strings in memory
- **Numeric literal**: number written in a program
  - No decimal point considered `int`, otherwise considered `float`
- Some operations behave differently depending on data type

Summary

- Typically applications perform 3 step process input - process - output
- Output to terminal standard out can be done with `print()` function
  - Functions are prewritten reusable code that have input arguments and can return a result
- Specify string literals using single `'` or double `"`
- Comment your code for human readers and maintainers
- A **variable** is a name that references a stored value in computer memory
  - assign (and reassign) values to variables
  - pass variables as arguments to function
  - variables have a type determine when assigned, `str` `int` `float`

References

Gaddis, T. (2021). *Starting out with python* (5th). Pearson.

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